Somak **Laha**

Ph.D. Candidate, Statistics · Harvard University

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Education

Harvard University *Cambridge, MA*

Ph.D. in Statistics | Advisor: Prof. Morgane Austern *Sept 2023 – Present*

Indian Statistical Institute *Kolkata, India*

M.Stat. in Statistics, specialization in Applied Statistics *Sept 2020 – June 2022*

Indian Statistical Institute *Kolkata, India*

B.Stat. (Honours) in Statistics *July 2017 – Sept 2020*

Industry Experience

Capital One Datalabs *Bangalore, India*

Data Science Analyst (Full-time) *July 2022 – Aug 2023*

- Built an ensemble statistical/ML model predicting used-car trade-in values for Capital One's AutoNavigator platform.
- Developed no-reference image quality assessment to gate car-image preprocessing, evaluating CLIP, SAM, MANIQA and YOLO.

Capital One Datalabs *Bangalore, India*

Data Analyst Intern *June – July 2021*

- Improved the design and interpretation of production A/B tests; implemented a Bayesian A/B testing widget in Python for analyst use.

Preprints

The Zero Pattern of a Design Matrix Drives Multiple Descent in Over-parameterized Regression *arXiv:2607.24041*

July 2026

K. H. Huang, H. Ye, **S. Laha**, M. Austern

- Work on over-parameterized regression almost always assumes the covariates are independent and their covariance is non-degenerate. Real design matrices are often neither. Structural zeros and collinear blocks are often the norm.
- We drop both assumptions and derive deterministic equivalents for the prediction risk in the propotional regime. The risk can be read off the variance profile of the design, and the result covers dependent covariates with singular covariance.
- Dependence and degeneracy replace the familiar double descent with multiple descent. We characterize precisely where the extra peaks sit, given the variance profile of the design.
- Maximum matchings and the Dulmage–Mendelsohn decomposition from graph theory help identify exactly which configurations make the variance singular, and those configurations are what locate the peaks.

Graph Attention Network for Node Regression on Random Geometric Graphs with Erdős–Rényi Contamination *arXiv:2601.23239*

Jan 2026

S. Laha, S. Liu, M. Austern | Submitted to **NeurIPS 2026**

- Graph attention networks cope well with noisy features and spurious edges in practice, but there is little theory showing that attention actually beats a plain graph neural network.
- We study node regression where the response depends on latent covariates but the observed covariates are noise-contaminated, and the graph is a random geometric graph on the latent covariates polluted by independent Erdős–Rényi edges. The geometric edges carry information about the latent positions; the contaminating edges carry none.
- Our designed attention layer scores neighbours so that the geometric ones dominate, builds denoised proxies for the latent covariates, and regresses on those. Cross-fitting keeps the proxy construction from biasing the regression that uses it.
- The proxies beat ordinary least squares on the noisy covariates for estimating the coefficient, and beat a vanilla GCN for predicting at an unlabelled node, both under mild growth conditions.

Research Experience and Academic Projects

Symmetry Breaking and Collective Bias in Populations of LLM Agents *Cambridge, MA*

With Prof. Subhabrata Sen, Department of Statistics, Harvard University *June 2026 – Present*

- Ask a single LLM which letter of the alphabet it prefers and it looks unbiased. Put a population of them in a naming game and one letter wins the convention far more often than chance. The suspicion is that a tiny bias, invisible in any one agent, is amplified by repeated communication.

- We model the population with Quantized Simplex Gossip, in which agents hold internal belief states and learn from one another's sampled outputs, so one agent's arbitrary draw becomes the next agent's evidence.
- We start the agents just barely off symmetry to investigate how small a bias can be amplified into the convention.

Depreferential Attachment Model through Preferential Attachment *Kolkata, India*

Advisor: Prof. Antar Bandyopadhyay, Statistics and Mathematics Unit, ISI *Jan 2022 – Present*

- Preferential attachment lets the largest vertices keep growing fastest, which is what produces power-law degrees. Many real systems push back against that.
- The model damps the leaders without inverting the attachment rule. Each step samples a set of old vertices in a degree-biased way, forbids them, then runs an ordinary preferential attachment step on whoever is left.
- Dissertation work: bounded the maximal degree under hard-core tabooing, and proved upper and lower bounds on the asymptotic order of a fixed vertex's degree and of the degree distribution under the soft-core scheme, with simulations alongside.
- Ongoing work studies fixed vertex degree and degree distribution asymptotics in sharper ways under varying taboo rate. It proves central limit theorems for fixed vertex degrees using martingale CLT techniques.
- Dissertation presentation

Edge-preserving Regression Surface Fitting *Kolkata, India*

Advisor: Prof. Partha Sarathi Mukherjee, ISRU, ISI | Group: Riddhiman Saha, Souhardya Ray *Aug 2020*

- Proposed a two-stage procedure that recovers discontinuous regression surfaces from noisy data. Neighborhood-based clustering finds the jumps, then local polynomial kernel smoothing fits around them.
- Project report & Presentation

Linear Models Projects: Shape Analysis and Satellite Image Stitching *Kolkata, India*

Advisor: Prof. Arnab Chakraborty, Applied Statistics Unit, ISI *Aug – Nov 2019*

- Fitted a Fourier descriptor model for convex shapes using linear mixed effects models, and wrote the fitting routine and an accompanying validity test in R.
- Estimated landmark positions from overlapping satellite screenshots via linear models and stitched them into a single vector map (R).

Teaching

Fall 2026	**Teaching Fellow**, STAT 211: Statistical Inference I (graduate)	*Harvard University*
Spring 2025	**Teaching Fellow**, STAT 212: Probability II (graduate)	*Harvard University*
Fall 2024	**Teaching Fellow**, STAT 120: Introduction to Bayesian Inference and Applications	*Harvard University*
	Led weekly sections, held office hours, and graded problem sets and exams.	

Talks and Presentations

Graph Attention Network for Node Regression on Random Geometric Graphs with Erdős–Rényi Contamination

Joint Statistical Meetings (JSM) 2026 *Boston, MA* · *Aug 2026*

2025 IMS International Conference on Statistics and Data Science (ICSDS) *Seville, Spain* · *Dec 2025*

Awards and Scholarships

2020	**Academic Award**, Prize for outstanding academic performance, 5th semester	*Indian Statistical Institute*
2017	**IIT-JEE Advanced**, All India Rank 676	*India*
2016	**KVPY Fellowship**, All India Rank 689, SX stream	*DST, Govt. of India*

Skills

Computing Python, R, C, SQL (Snowflake)